CLMS to Copernicus Data Space Ecosystem Migration

Weekly Status Update

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```
// ── Hide Quarto code-fold disclosure triangles ──
hideCode = new Promise(r ⇒ { if (document.readyState ≡≡ "complete") setTimeout(r, 200); else
addEventListener("load", () ⇒ setTimeout(r, 500)); }).then(() ⇒ {
                          document.querySelectorAll("details.quarto-code-fold,             .cell-
code, .sourceCode").forEach(function(el) { el.style.display = "none"; });
})

// ── Load data from external URL (not from repo) ──
data  =  await  fetch("https://raw.githubusercontent.com/copernicus-land/clms-cdse-migration-data/
main/migration_data.json")
  .then(r ⇒ r.json())
  .catch(e ⇒ { console.error("Failed to load data", e); return {groups:[], portfolio:{}, cdse:
{}, summary:{}} })
groups = data.groups
summary = data.summary
portfolio = data.portfolio
cdse = data.cdse
lastUpdated = data.scraped_at.slice(0,10)

// ── Load daily OData file tracking ──
odataDaily = await fetch("https://raw.githubusercontent.com/copernicus-land/clms-cdse-migration-
data/main/odata_daily.json")
  .then(r ⇒ r.json())
  .catch(e ⇒ { console.error("Failed to load odata_daily", e); return {history: []} })

// ── Filter groups ──
componentOf = d ⇒ {
  if (d.cdse_datasets?.length > 0) return d.cdse_datasets[0].component
  return d.lp_datasets?.[0]?.component || ""
}
groupHasComponent = (g, comp) ⇒ {
  if (g.cdse_datasets?.some(d ⇒ d.component ≡≡ comp)) return true
  if (g.lp_datasets?.some(d ⇒ d.component ≡≡ comp)) return true
  return false
}
filtered = groups.filter(d ⇒ {
  let matches = false
  for (const comp of activeComponents) {
    if (groupHasComponent(d, comp)) { matches = true; break }
  }
  if (!matches) return false
  if (searchTerm && !d.name.toLowerCase().includes(searchTerm.toLowerCase())) return false
  return true
})

// ── Aggregate by category ──
categories = [...new Set(filtered.map(d ⇒ d.category).filter(Boolean))].sort()

// ── Totals reflecting the current filter selection ──
filteredTotals = ({
  datasets: d3.sum(filtered, d ⇒ d.total),
  groups: filtered.length
})
catStats = categories.map(cat ⇒ {
  const g = filtered.filter(d ⇒ d.category ≡≡ cat)
```

```
  return {
    name: cat,
    groups: g,
    total: d3.sum(g, d ⇒ d.total),
    on_cdse: d3.sum(g, d ⇒ d.on_cdse),
    pct: Math.round(d3.sum(g, d ⇒ d.on_cdse) / Math.max(d3.sum(g, d ⇒ d.total), 1) * 100),
    s3: d3.sum(g, d ⇒ d.s3),
    stac: d3.sum(g, d ⇒ d.stac),
    openeo: d3.sum(g, d ⇒ d.openeo),
    shub: d3.sum(g, d ⇒ d.shub),
  }
})

// ── Helpers ──
pctColor = (pct) ⇒ pct ⩾ 100 ? "#00aa00" : pct ⩾ 75 ? "#66cc00" : pct ⩾ 50 ? "#ffcc00" : pct
⩾ 25 ? "#ff8800" : "#ff0000"

pctBar = (pct) ⇒ {
  const capped = Math.min(pct, 100)
  const color = pctColor(pct)
  const filled = "▉".repeat(Math.floor(capped / 20))
  const empty = "▒".repeat(Math.max(0, 5 - Math.floor(capped / 20)))
      return    htl.html`<span    style="color:${color};font-weight:bold">${pct}%</span>    <span
style="color:#bbb">${filled}${empty}</span>`
}

badge = (val, total) ⇒ {
  if (val ⩵ 0 && total ⩵ 0) return htl.html`<span style="background:#888;color:#fff;padding:1px
6px;border-radius:3px;font-weight:bold">–</span>`
  const pct = Math.min(val / Math.max(total, 1), 1)
  const color = pct ⩾ 1 ? "#00aa00" : pct ⩾ 0.75 ? "#66cc00" : pct ⩾ 0.5 ? "#ffcc00" : pct ⩾
0.25 ? "#ff8800" : "#ff0000"
      return    htl.html`<span    style="background:${color};color:#fff;padding:1px    6px;border-
radius:3px;font-weight:bold">${val}</span>`
}

check = (on) ⇒ on
  ? htl.html`<span style="color:#00aa00;font-weight:bold">✓</span>`
  : htl.html`<span style="color:#ccc">×</span>`

// ── Changes since last week ──
changes = data.changes_since_last_week || []
newGroups = changes.filter(d ⇒ d.newly_on_cdse)
positiveChanges = changes.filter(d ⇒ d.s3 > 0 && !d.newly_on_cdse)
lastWeek = changes.length > 0 ? "since " + data.scraped_at.slice(0,10) : ""
odataDelta = data.odata_delta || 0
dailyDelta = data.daily_delta || 0
weeklyDelta = data.weekly_delta || 0
odataDailyAvg = data.odata_daily_avg || 0

// ── Toggle functions for expand/collapse (plain DOM, no OJS reactivity) ──
toggleScript = {
  window.toggleCategory = function(catName) {
    const key = catName.replace(/"/g, '&amp;quot;');
      const groupRows = document.querySelectorAll('tr[data-category="' + key + '"]:not([data-
```

```
group])');
    const isExpanded = groupRows.length > 0 && groupRows[0].style.display !== 'none';
    groupRows.forEach(function(r) { r.style.display = isExpanded ? 'none' : ''; });
    if (isExpanded) {
      // Collapsing the category also collapses any open dataset rows within it
     document.querySelectorAll('tr[data-category="' + key + '"][data-group]').forEach(function(r)
{ r.style.display = 'none'; });
       document.querySelectorAll('span[data-group-arrow^="' + key + '::"]').forEach(function(a)
{ a.textContent = '▶'; });
    }
    const arrow = document.querySelector('span[data-arrow="' + key + '"]');
    if (arrow) arrow.textContent = isExpanded ? '▶' : '▼';
  }
  window.toggleGroup = function(groupKey) {
    const key = groupKey.replace(/"/g, '&quot;');
    const rows = document.querySelectorAll('tr[data-group="' + key + '"]');
    const isExpanded = rows.length > 0 && rows[0].style.display !== 'none';
    rows.forEach(function(r) { r.style.display = isExpanded ? 'none' : ''; });
    const arrow = document.querySelector('span[data-group-arrow="' + key + '"]');
    if (arrow) arrow.textContent = isExpanded ? '▶' : '▼';
  }
}
```

```
// ── Onboarding history plot ──

onboardingPlot = {
  const barHeight = 28
  let mode = "daily"

  function getWeekNumber(dateStr) {
    const d = new Date(dateStr)
    const start = new Date(d.getFullYear(), 0, 1)
    const days = Math.floor((d - start) / 86400000)
    return Math.ceil((days + start.getDay() + 1) / 7)
  }

  function buildChart(container) {
    container.innerHTML = ""

    const data = mode === "daily"
       ? (odataDaily.history || []).slice().reverse().map(h => ({label: h.date, value: h.delta,
color: h.delta >= 0 ? '#e67e22' : '#e74c3c'}))
      : [{label: "WoY " + getWeekNumber(odataDaily.scraped_at), value: odataDaily.weekly_delta ||
0, color: '#e67e22'}]

    const maxVal = Math.max(...data.map(d => Math.abs(d.value)), 1)

    data.forEach(d => {
      const pct = Math.abs(d.value) / maxVal
      const row = document.createElement("div")
      row.style.cssText = "display:flex;align-items:center;margin:2px 0;font-size:0.85em"

      const label = document.createElement("span")
      label.textContent = d.label
```

```
          label.style.cssText  =  "width:100px;text-align:right;padding-right:8px;color:#555;flex-
shrink:0"
      row.appendChild(label)

      const track = document.createElement("div")
      track.style.cssText = `height:${barHeight}px;flex:1;display:flex;align-items:center`

      const fill = document.createElement("div")
                  fill.style.cssText   =   `height:${barHeight   -   4}px;width:${Math.max(pct
*   100,   2)}%;background:${d.color};border-radius:3px;display:flex;align-items:center;padding-
left:6px;min-width:fit-content`
      fill.textContent = (d.value ⩾ 0 ? "+" : "") + Math.abs(d.value).toLocaleString()
      fill.style.color = "#fff"
      fill.style.fontSize = "0.85em"
      fill.style.fontWeight = "bold"
      track.appendChild(fill)
      row.appendChild(track)
      container.appendChild(row)
    })

    // Radio buttons below the chart
    const controls = document.createElement("div")
    controls.style.cssText = "display:flex;gap:16px;margin-top:8px;align-items:center"

    ;["daily", "weekly"].forEach(m ⇒ {
      const lbl = document.createElement("label")
                          lbl.style.cssText    =    "display:flex;align-items:center;gap:4px;font-
size:0.85em;cursor:pointer;color:#555"

      const radio = document.createElement("input")
      radio.type = "radio"
      radio.name = "onboardingMode"
      radio.value = m
      radio.checked = m ⩵ mode
      radio.style.cursor = "pointer"

      radio.addEventListener("change", () ⇒ {
        mode = m
        buildChart(container)
      })

      lbl.appendChild(radio)
      lbl.append(m)
      controls.appendChild(lbl)
    })

    container.appendChild(controls)
  }

  const container = document.createElement("div")
  container.style.marginTop = "8px"
  buildChart(container)
  return container
}
```

1. Migration History

```
html`
${newGroups.length > 0 ? htl.html `
<div            style="background:#e8f5e9;border:1px            solid            #a5d6a7;border-
radius:6px;padding:12px;margin:12px 0">
  <strong>🆕 Products ${lastWeek}</strong>
  <div style="margin:4px 0"><strong>Newly onboarded:</strong> ${newGroups.map(d ⇒ d.name).join(",
")}</div>
</div>` : ""}

${positiveChanges.length > 0 ? htl.html `
<div  style="background:#e8f5e9;border:1px  solid  #a5d6a7;border-radius:6px;padding:12px  12px  4px
12px;margin:12px 0">
  <strong>🆕 Products ${lastWeek}</strong>
   ${positiveChanges.map(d ⇒ htl.html `<div style="margin:4px 0">${d.name}: <strong>+${d.s3}</
strong> S3, <strong>+${d.stac}</strong> STAC, <strong>+${d.openeo}</strong> OpenEO</div>`)}
</div>` : ""}

${changes.length ≡≡ 0 ? htl.html `<div style="background:#f5f5f5;border:1px solid #ddd;border-
radius:6px;padding:12px;margin:12px 0;color:#888">No new CLMS product were onboarded this week.</
div>` : ""}

${(odataDaily.weekly_delta || odataDaily.daily_delta || (odataDaily.history || []).length > 0) ?
htl.html `
<div            style="background:#fff3e0;border:1px            solid            #ffcc80;border-
radius:6px;padding:12px;margin:12px 0">
  <strong>📦 Onboarding history</strong>
  ${odataDaily.daily_avg ? htl.html `<div style="margin:4px 0;color:#888;font-size:0.85em">Daily
average: ${odataDaily.daily_avg.toLocaleString()} new files/day</div>` : ""}
  ${onboardingPlot}
</div>` : ""}
`
```

2. Upcoming

Details to be provided — placeholder for upcoming CLMS products planned for CDSE ingestion.

Preparing dataset descriptions, expected timelines, and service coverage for the next batch of products to be onboarded.

---

3. CLMS Product on CDSE

This table shows the full CLMS portfolio with their migration status to the CDSE.

```
// Toggle-button style filter: click a component to enable/disable it
// (multi-select, all enabled by default). Replaces the old single-select
// dropdown, which only let one component be viewed at a time.
viewof activeComponents = {
```

```
  const allComponents = [...new Set(groups.map(d => componentOf(d)).filter(Boolean))].sort()
  const active = new Set(allComponents)

  const container = document.createElement("div")
     container.style.cssText   =   "display:flex;gap:8px;flex-wrap:wrap;align-items:center;margin-
bottom:12px";

  function render() {
    container.innerHTML = "";
    allComponents.forEach(comp => {
      const isActive = active.has(comp);
      const btn = document.createElement("button");
      btn.type = "button";
      btn.textContent = comp;
               btn.style.cssText   =   `padding:6px   14px;border-radius:16px;cursor:pointer;font-
size:0.85em;border:1px solid ${isActive ? "#0d6efd" : "#ccc"};background:${isActive ? "#0d6efd" :
"#f5f5f5"};color:${isActive ? "#fff" : "#555"}`;
      btn.addEventListener("click", () => {
        if (active.has(comp)) active.delete(comp);
        else active.add(comp);
        render();
        container.value = new Set(active);
        container.dispatchEvent(new CustomEvent("input"));
      });
      container.appendChild(btn);
    });
  }
  render();
  container.value = new Set(active);
  return container;
}

viewof searchTerm = Inputs.text({label: "Search", placeholder: "Product group…", width: 250})
```

```
// ── Build the table via plain DOM APIs ──
// so that column widths defined on <colgroup> are guaranteed to apply to
// every row, including rows toggled between hidden/visible.
productGroupsTable = {
  const COLS = 8;
  const colWidths = ["auto", "9%", "9%", "9%", "9%", "9%", "9%", "18%"];

  function makeCell(tag, html, opts = {}) {
    const el = document.createElement(tag);
    if (html instanceof Node) el.appendChild(html);
    else el.innerHTML = html;
    el.style.padding = opts.padding ?? "6px";
    el.style.whiteSpace = "nowrap";
    el.style.overflow = "hidden";
    el.style.textOverflow = "ellipsis";
    if (opts.center) el.style.textAlign = "center";
    if (opts.bold) el.style.fontWeight = "bold";
    if (opts.title) el.title = opts.title;
    if (opts.style) el.style.cssText += ";" + opts.style;
    return el;
```

```
  }

  const wrap = document.createElement("div");

  const table = document.createElement("table");
  table.style.cssText = "width:100%;border-collapse:collapse;font-size:0.8em;table-layout:fixed";

  const colgroup = document.createElement("colgroup");
  colWidths.forEach(w => {
    const col = document.createElement("col");
    col.style.width = w;
    colgroup.appendChild(col);
  });
  table.appendChild(colgroup);

  const thead = document.createElement("thead");
  const headRow = document.createElement("tr");
  headRow.style.cssText = "background:#f0f0f0;text-align:left";
  ["Category", "Total", "CDSE", "S3", "STAC", "OpenEO", "S.Hub", "Progress"].forEach((label, i)
=> {
    const th = makeCell("th", label, { center: i > 0 && i < 7, padding: "6px" });
    th.style.borderBottom = "2px solid #ddd";
    headRow.appendChild(th);
  });
  thead.appendChild(headRow);
  table.appendChild(thead);

  const tbody = document.createElement("tbody");

  catStats.forEach(cat => {
    const tr = document.createElement("tr");
    tr.style.cssText = "background:#f5f5f5;border-bottom:1px solid #ccc;cursor:pointer";

    const nameCell = makeCell("td", "", { bold: true, title: cat.name });
    const arrow = document.createElement("span");
    arrow.dataset.arrow = cat.name;
    arrow.textContent = "▶";
    nameCell.appendChild(arrow);
    nameCell.append(" " + cat.name);
    tr.appendChild(nameCell);

    tr.appendChild(makeCell("td", String(cat.total), { center: true, bold: true }));
    tr.appendChild(makeCell("td", String(cat.on_cdse), { center: true, bold: true }));
    tr.appendChild(makeCell("td", badge(cat.s3, cat.total), { center: true }));
    tr.appendChild(makeCell("td", badge(cat.stac, cat.total), { center: true }));
    tr.appendChild(makeCell("td", badge(cat.openeo, cat.total), { center: true }));
    tr.appendChild(makeCell("td", badge(cat.shub, cat.total), { center: true }));
    tr.appendChild(makeCell("td", pctBar(cat.pct), {}));

    tr.addEventListener("click", () => window.toggleCategory(cat.name));
    tbody.appendChild(tr);

    cat.groups.forEach(g => {
      const groupKey = cat.name + "::" + g.name;
      const gtr = document.createElement("tr");
```

```
      gtr.dataset.category = cat.name;
      gtr.style.cssText = "border-bottom:1px solid #ddd;display:none;cursor:pointer";

      const gNameCell = makeCell("td", "", { title: g.name, padding: "4px 6px 4px 24px" });
     const dsInFilter = (g.cdse_datasets || []).filter(ds ⇒ activeComponents.has(ds.component));
               const   lpOffCdse   =   (g.lp_datasets   ||   []).filter(lp   ⇒   !lp.on_cdse   &&
activeComponents.has(lp.component));
      const hasDatasets = dsInFilter.length + lpOffCdse.length > 0;
      if (hasDatasets) {
        const gArrow = document.createElement("span");
        gArrow.dataset.groupArrow = groupKey;
        gArrow.textContent = "▶";
        gArrow.style.marginRight = "4px";
        gNameCell.appendChild(gArrow);
      }
      const link = document.createElement("a");
      link.href = g.url;
      link.target = "_blank";
      link.textContent = g.name;
      link.addEventListener("click", (e) ⇒ e.stopPropagation());
      gNameCell.appendChild(link);
      gtr.appendChild(gNameCell);

       gtr.appendChild(makeCell("td", String(g.total), { center: true, bold: true, padding: "4px
6px" }));
        gtr.appendChild(makeCell("td", String(g.on_cdse), { center: true, bold: true, padding:
"4px 6px" }));
     gtr.appendChild(makeCell("td", badge(g.s3, g.total), { center: true, padding: "4px 6px" }));
    gtr.appendChild(makeCell("td", badge(g.stac, g.total), { center: true, padding: "4px 6px" }));
        gtr.appendChild(makeCell("td", badge(g.openeo, g.total), { center: true, padding: "4px
6px" }));
    gtr.appendChild(makeCell("td", badge(g.shub, g.total), { center: true, padding: "4px 6px" }));
      gtr.appendChild(makeCell("td", pctBar(g.pct), { padding: "4px 6px" }));

      if (hasDatasets) {
        gtr.addEventListener("click", () ⇒ window.toggleGroup(groupKey));
      }
      tbody.appendChild(gtr);

      dsInFilter.forEach(ds ⇒ {
        const dtr = document.createElement("tr");
        dtr.dataset.category = cat.name;
        dtr.dataset.group = groupKey;
        dtr.style.cssText = "border-bottom:1px solid #eee;display:none;background:#fafafa";

        dtr.appendChild(makeCell("td", ds.id, { padding: "3px 6px 3px 44px", title: ds.id }));
        dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));
        dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));
        dtr.appendChild(makeCell("td", check(ds.s3), { center: true, padding: "3px 6px" }));
        dtr.appendChild(makeCell("td", check(ds.stac), { center: true, padding: "3px 6px" }));
        dtr.appendChild(makeCell("td", check(ds.openeo), { center: true, padding: "3px 6px" }));
        dtr.appendChild(makeCell("td", check(ds.shub), { center: true, padding: "3px 6px" }));
        dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));

        tbody.appendChild(dtr);
```

```
      });

      lpOffCdse.forEach(lp => {
        const dtr = document.createElement("tr");
        dtr.dataset.category = cat.name;
        dtr.dataset.group = groupKey;
        dtr.style.cssText = "border-bottom:1px solid #eee;display:none;background:#fafafa";

        dtr.appendChild(makeCell("td", lp.title, { padding: "3px 6px 3px 44px", title: lp.title,
style: "color:#999;font-style:italic" }));
        dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));
        dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));
             dtr.appendChild(makeCell("td", "—", { center: true, padding: "3px 6px", style:
"color:#ccc" }));
             dtr.appendChild(makeCell("td", "—", { center: true, padding: "3px 6px", style:
"color:#ccc" }));
             dtr.appendChild(makeCell("td", "—", { center: true, padding: "3px 6px", style:
"color:#ccc" }));
             dtr.appendChild(makeCell("td", "—", { center: true, padding: "3px 6px", style:
"color:#ccc" }));
        dtr.appendChild(makeCell("td", "", { padding: "3px 6px" }));

        tbody.appendChild(dtr);
      });
    });
  });

  // ── Summary row (col sums over the currently filtered groups) ──
  const sumRow = document.createElement("tr");
  sumRow.style.cssText = "background:#e0e0e0;border-top:2px solid #999;font-weight:bold";
  const sumName = makeCell("td", "Total", { bold: true });
  sumName.style.fontStyle = "italic";
  sumRow.appendChild(sumName);

  const sumTotal = d3.sum(catStats, d => d.total);
  const sumCDSE = d3.sum(catStats, d => d.on_cdse);
  const sumS3 = d3.sum(catStats, d => d.s3);
  const sumSTAC = d3.sum(catStats, d => d.stac);
  const sumOEO = d3.sum(catStats, d => d.openeo);
  const sumSHub = d3.sum(catStats, d => d.shub);
  const sumPct = Math.round(sumCDSE / Math.max(sumTotal, 1) * 100);

  sumRow.appendChild(makeCell("td", String(sumTotal), { center: true, bold: true }));
  sumRow.appendChild(makeCell("td", String(sumCDSE), { center: true, bold: true }));
  sumRow.appendChild(makeCell("td", badge(sumS3, sumTotal), { center: true }));
  sumRow.appendChild(makeCell("td", badge(sumSTAC, sumTotal), { center: true }));
  sumRow.appendChild(makeCell("td", badge(sumOEO, sumTotal), { center: true }));
  sumRow.appendChild(makeCell("td", badge(sumSHub, sumTotal), { center: true }));
  sumRow.appendChild(makeCell("td", pctBar(sumPct), {}));
  tbody.appendChild(sumRow);

  table.appendChild(tbody);
  wrap.appendChild(table);
  return wrap;
}
```

4. Services

- S3 CSV Catalogue
- OData API
- STAC Browser
- OpenEO
- Sentinel Hub
- CDSE Documentation

5. Methodology

Per-service counts are matched by:

- **S3**: Direct catalogue count per product type
- **STAC**: Fuzz-matched by checking if CDSE dataset ID (minus `_cog`/`_nc` suffix) appears in STAC collection IDs
- **OpenEO**: Matched by constructing `CLMS_{id.upper()}` convention
- **Sentinel Hub**: Estimated from S3 availability